R&D System User Manual

Overview

The R&D system manages the company's drink and food recipes and calculates the cost per cup in real time from the material prices approved by Finance. Open it from the office Company Portal via "R&D" (URL /rd). It has four pages; the buttons at the top switch between them.

Page	URL	What it does
IMT ERP R&D	/rd	Recipes, store cost per cup, price list, product prices, POS product mapping, usage variance check
R&D Brain (Lab)	/rd-lab	Structured recipes, ingredient library, recipe calculator and sugar tax check, experiments, competitors and additives, external knowledge, 6 AI agents
R&D Discussion AI	/rd-ai	Discuss recipes and R&D questions with AI; add your own knowledge
R&D Logs	/rd-logs	R&D logs, added by hand, bulk imported, or summarised automatically from R&D Discussion AI chats

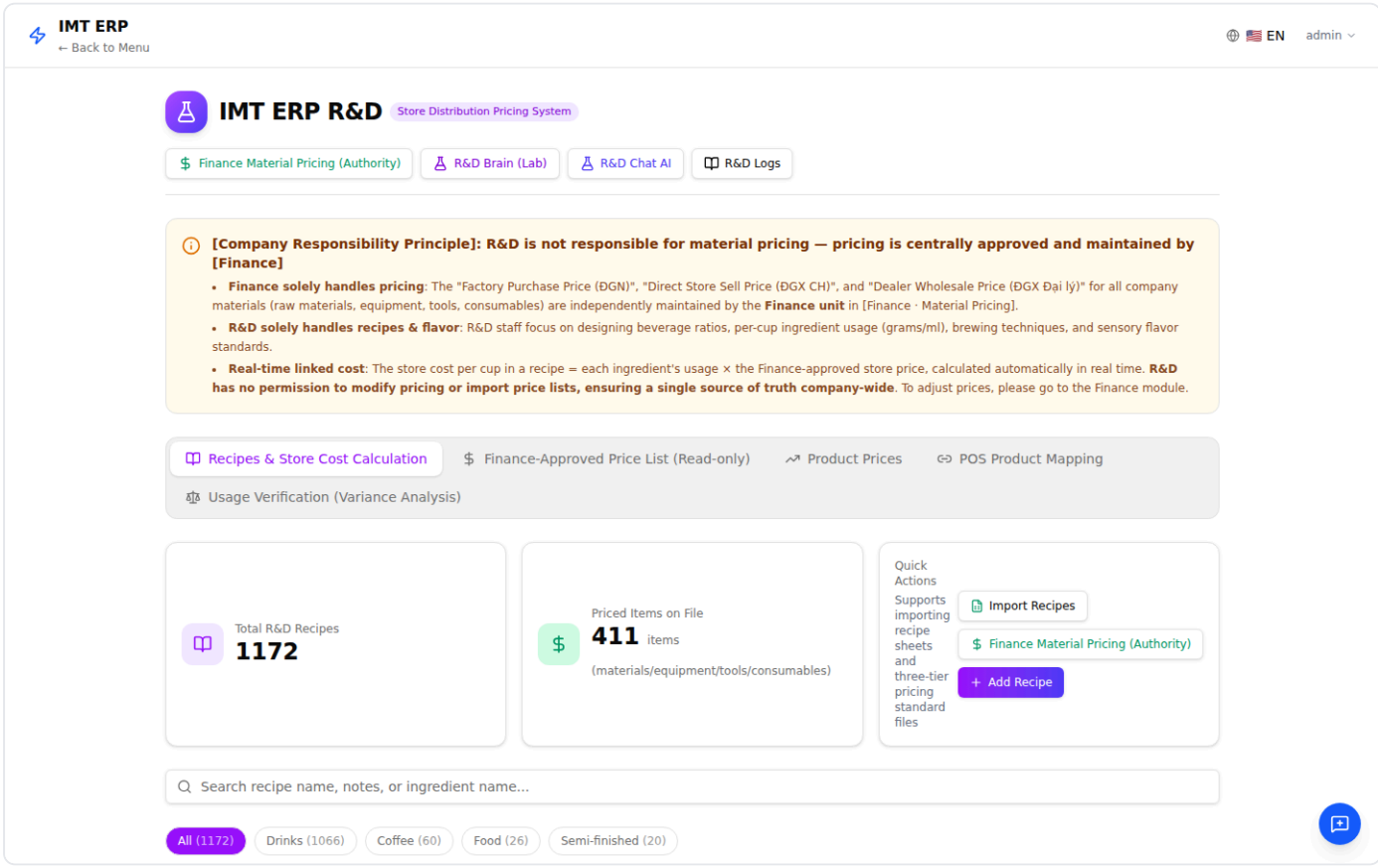
Access

1. Only members of the R&D unit can open it; anyone else sees "Only R&D unit can access the R&D module".
2. Switch between Traditional Chinese, English and Vietnamese at the top right; AI replies use the language you pick.

Division of responsibilities

R&D does not set material prices. The factory purchase price (ĐGN), direct store price (ĐGX CH) and dealer wholesale price (ĐGX Đại lý) of every material (ingredients, equipment, tools, consumables) are maintained by Finance under "Finance Material Pricing". R&D is responsible for recipe ratios, quantity per cup, preparation method and flavour standards. A recipe's store cost per cup = each ingredient's quantity × the store price approved by Finance. The system calculates it automatically, and R&D cannot change unit prices.

Recipes & Store Cost Calculation

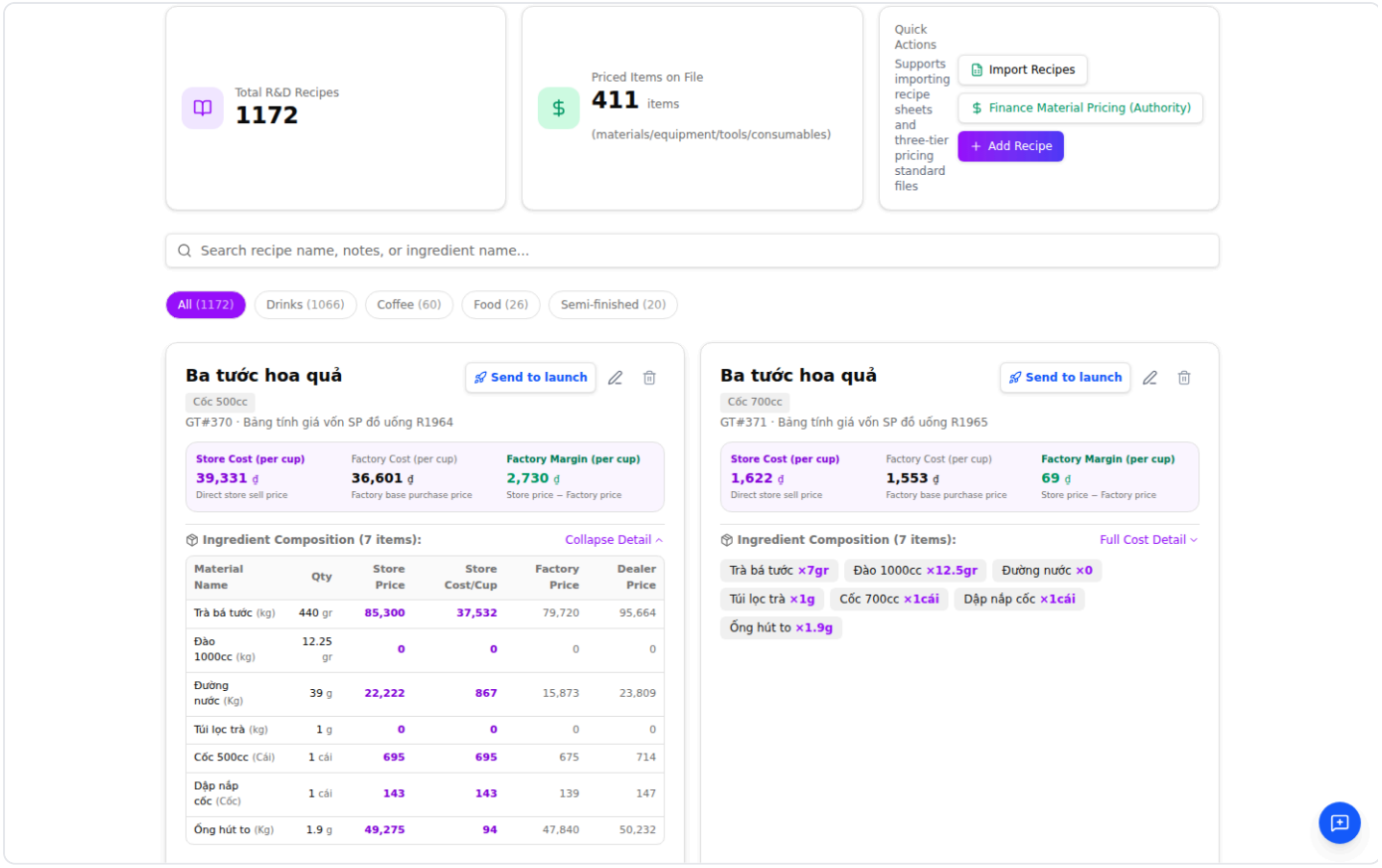


The cards at the top show the total number of R&D recipes and the number of priced items on file. Below, filter by All, Drinks, Coffee, Food or Semi-finished, or search by recipe name, notes, ingredient name or code.

Each recipe card shows three figures:

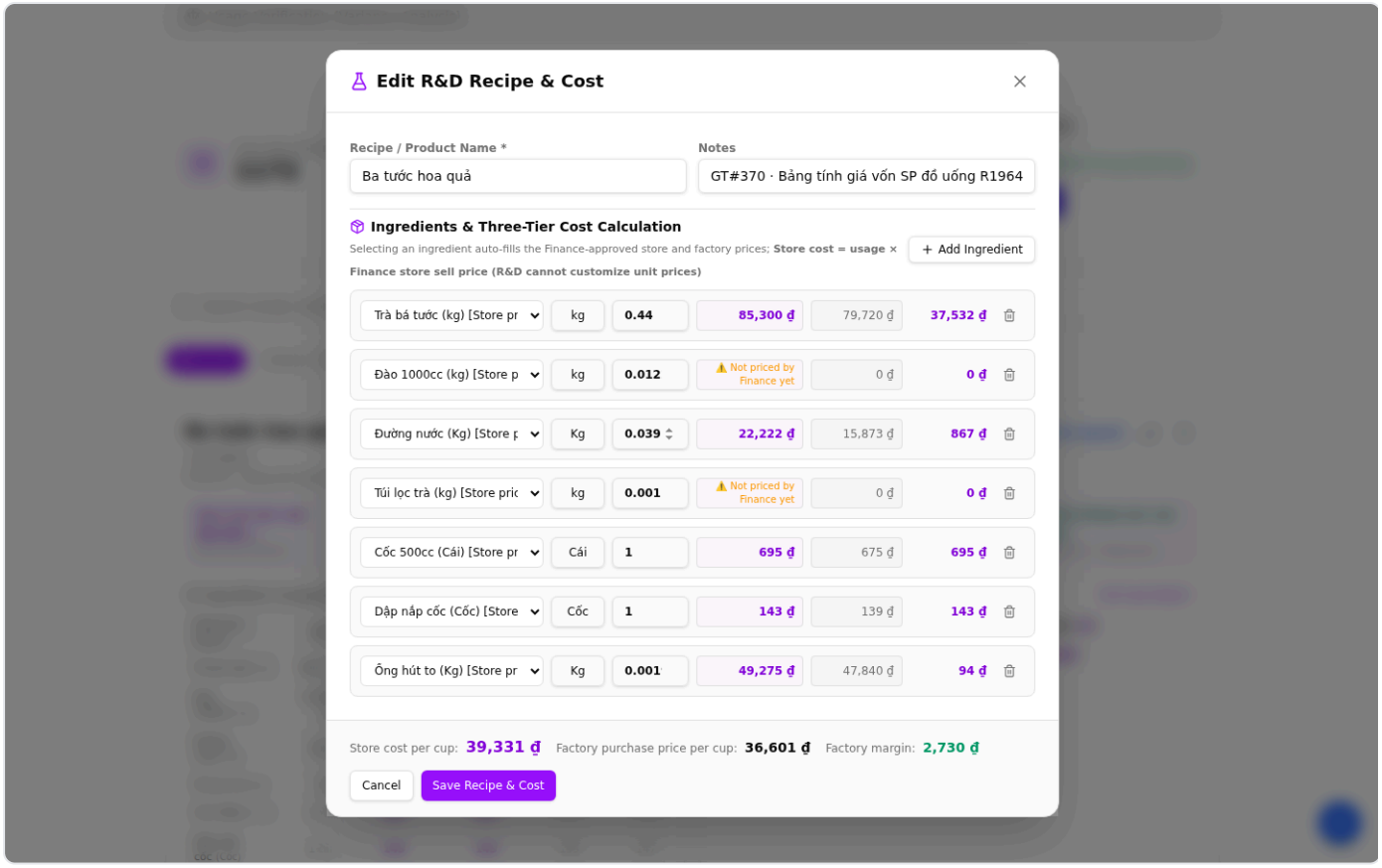
Figure	How it is calculated
Store Cost (per cup)	Each ingredient's quantity × direct store price
Factory Cost (per cup)	Each ingredient's quantity × factory purchase price
Factory Margin (per cup)	Store cost − factory cost

Click "Full Cost Detail" to see each ingredient's quantity, direct store price, store cost per cup, factory price and dealer price.



Add or edit a recipe

1. Click "Add Recipe", or the pencil icon on a recipe card to edit it.
2. Fill in "Recipe / Product Name" (required) and "Notes".
3. Click "Add Ingredient" and pick an ingredient from the list. The store price and factory price approved by Finance fill in automatically.
4. Enter the quantity per cup. "⚠️ Not priced by Finance yet" next to an ingredient means Finance has not priced it, so it counts as 0. Ask Finance to add the price.
5. The bottom of the window shows the store cost per cup, factory cost per cup and factory margin as you type.
6. Click "Save Recipe & Cost".



Import recipes

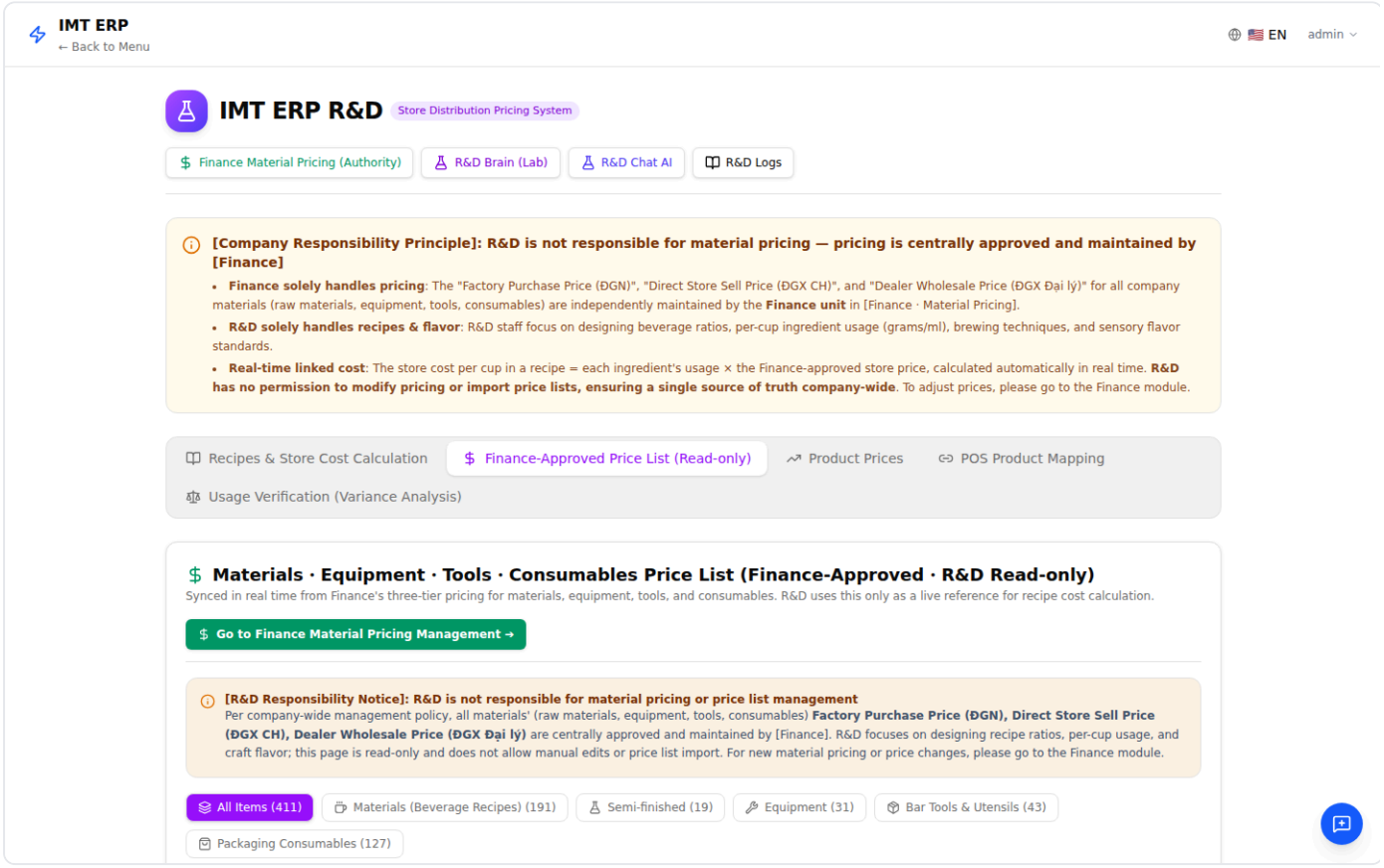
Click "Import Recipes" and upload an Excel file (.xlsx / .xls). The system reads the first sheet and matches these columns: recipe name (or product name), notes, ingredient code, ingredient name, quantity per cup. Recipes with the same name are updated.

Send to Marketing

When a recipe is final, click "Send to launch" on its card. The recipe goes to the Marketing system's new product launch pipeline, where Marketing prepares the VIP early-access period and launch schedule. The default price sent is store cost × 2.5; Marketing can change it.

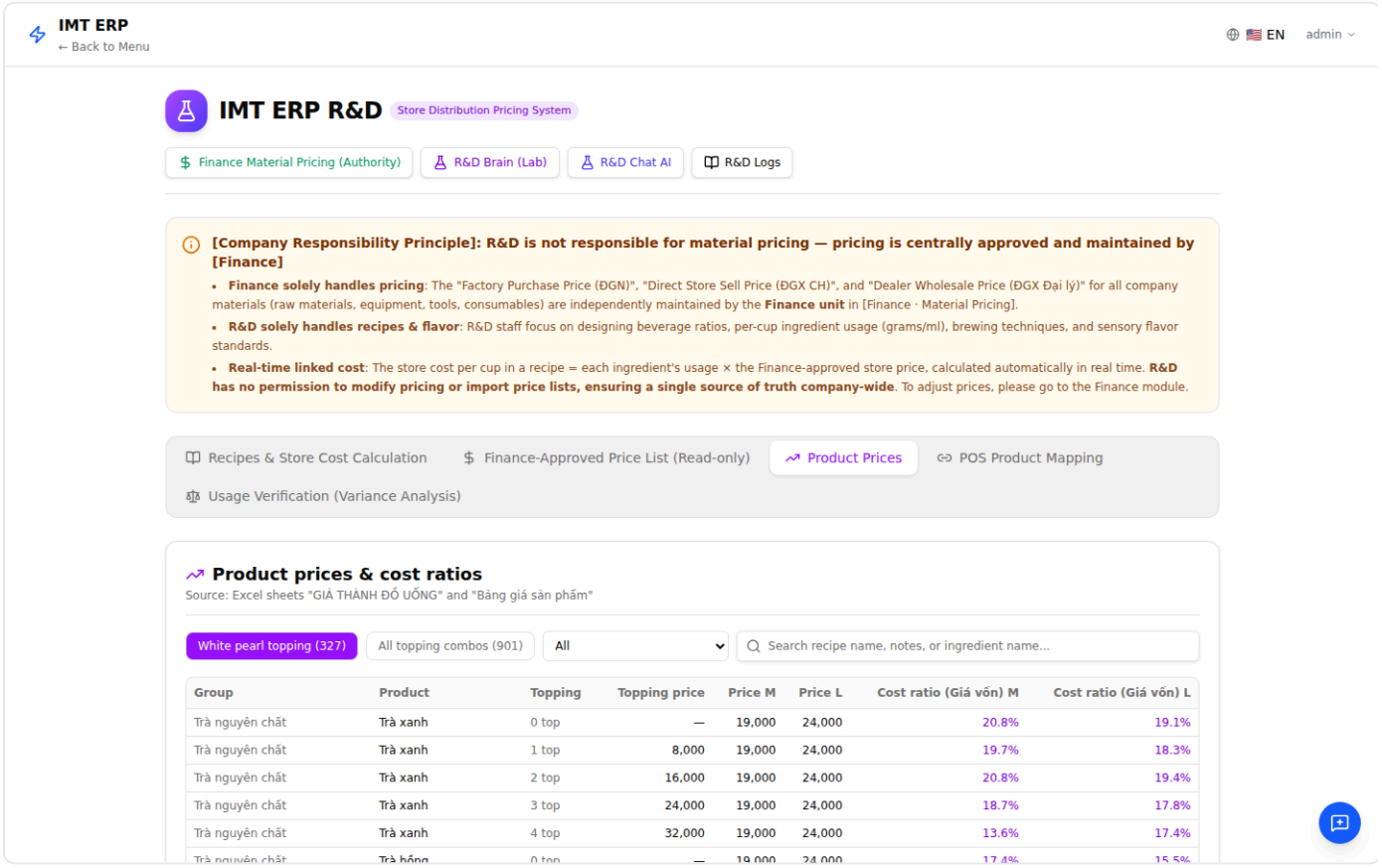
Price list, product prices and POS mapping

Finance-Approved Price List (Read-only)



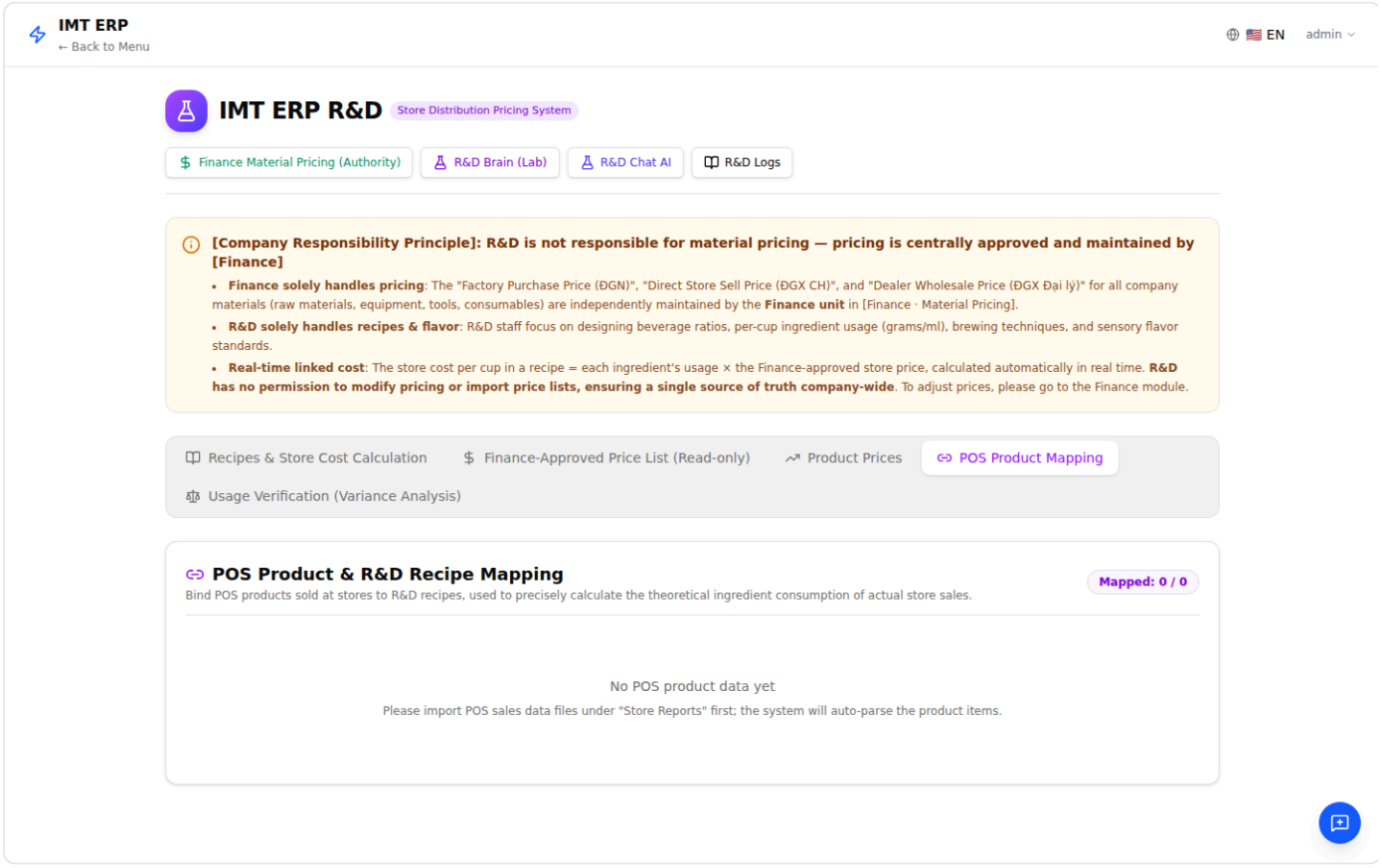
Shows Finance's three price levels in real time. Filter by category (materials, semi-finished, equipment, bar tools, packaging consumables) or search. R&D can only view it. To change a price, click "Go to Finance Material Pricing Management" or contact Finance.

Product Prices



Product prices, topping prices and cost ratios taken from the Excel sheets "GIÁ THÀNH ĐỒ UỐNG" and "Bảng giá sản phẩm". Switch between "White pearl topping" and "All topping combos", and search by item.

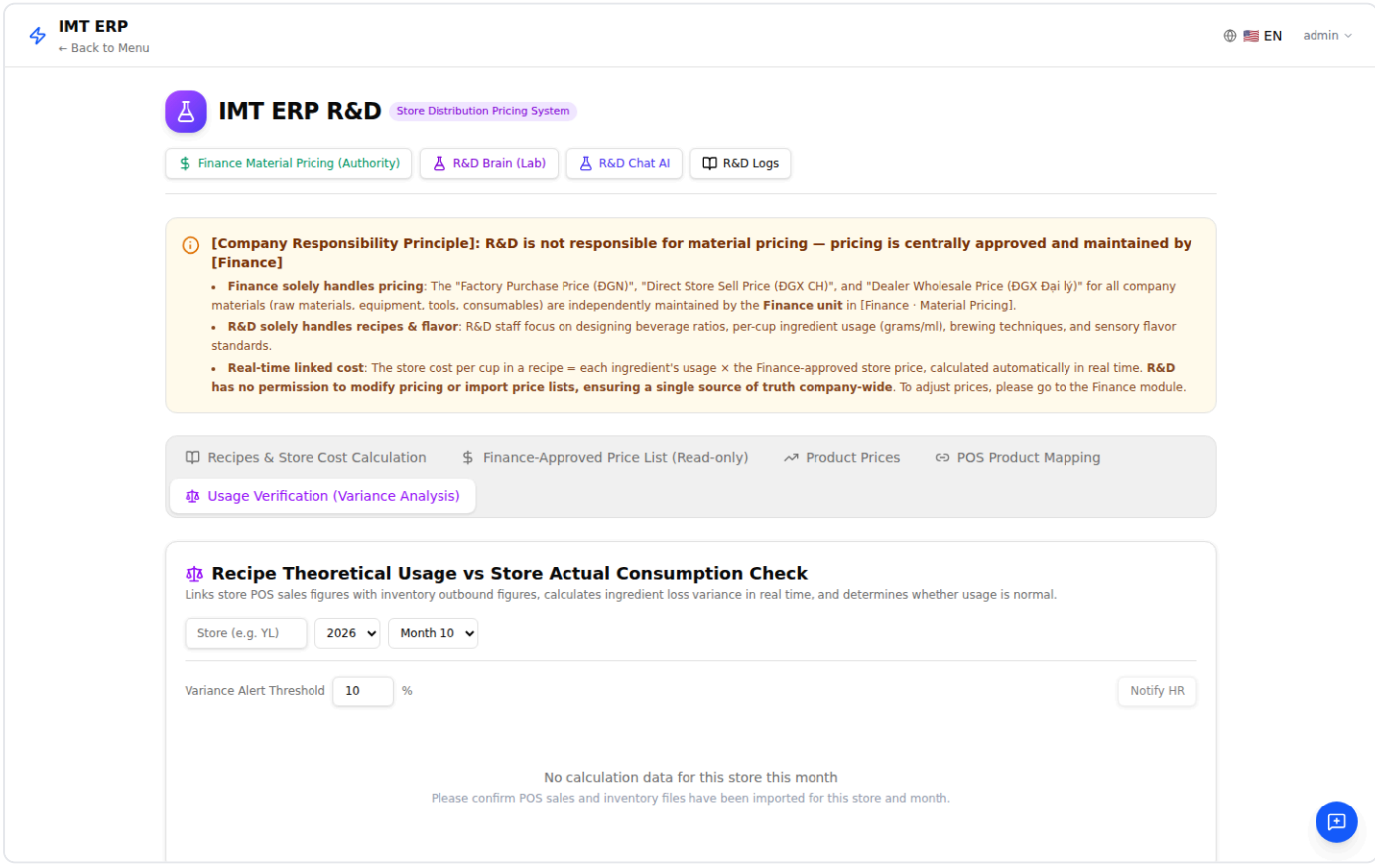
POS Product Mapping



Link each product sold through the store POS to an R&D recipe, so the system can work out how much of each ingredient sales should have used.

1. Import the POS sales data in "Store Reports" first. The system extracts the products from it.
2. Come back here and pick the matching recipe for each POS product under "Mapped R&D Recipe".
3. The top right shows how many are mapped. Unmapped products are left out of theoretical usage.

Usage Verification (Variance Analysis)



Compares theoretical usage (units sold × recipe quantity) with actual stock issued, to find ingredients with abnormal waste.

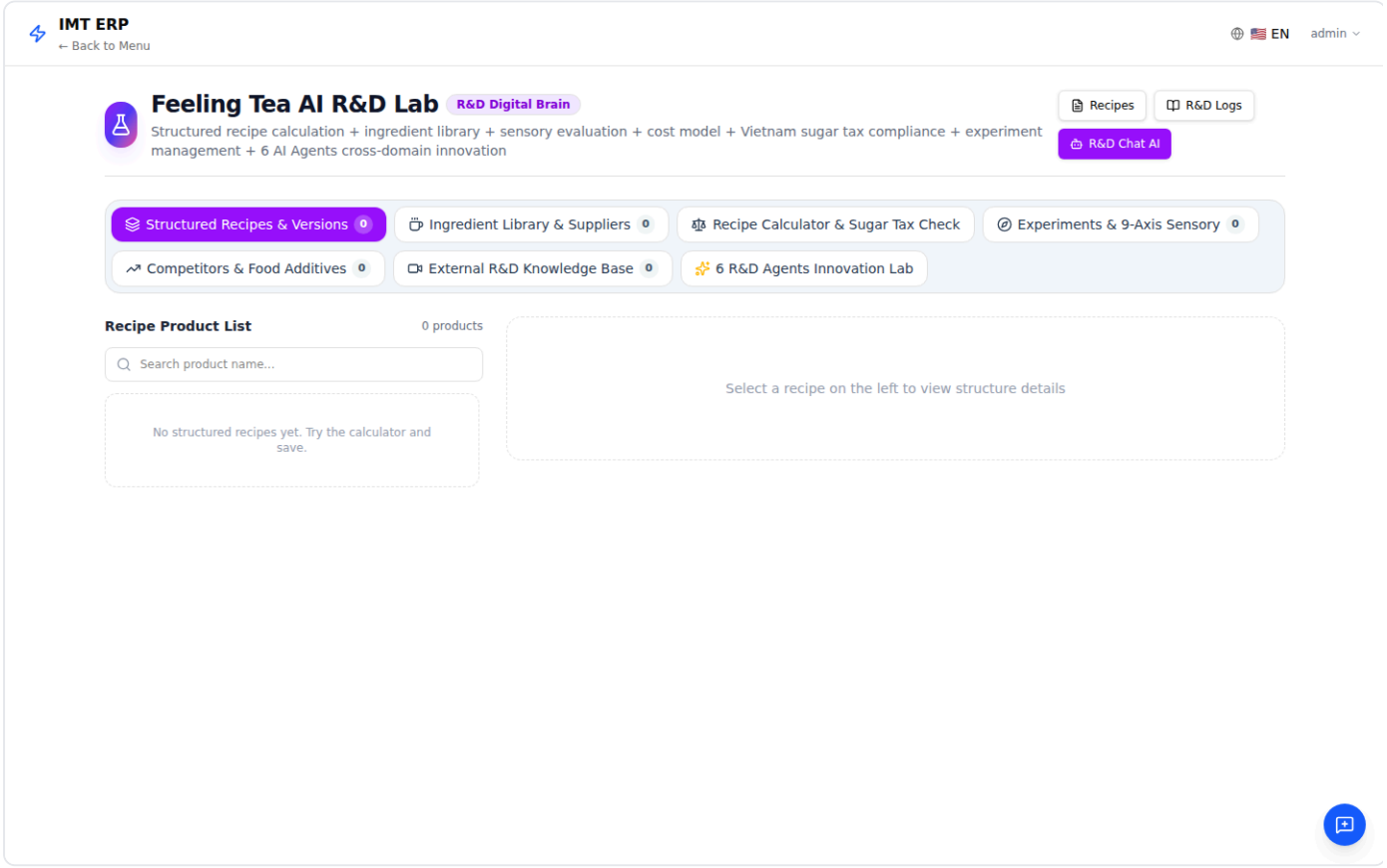
1. Choose the store, year and month. POS sales and inventory data for that store and month must be imported first.
2. Set the "Variance Alert Threshold" (%). Ingredients above it are marked "Over Threshold".
3. The table lists each ingredient's theoretical usage, actual stock issued, difference, variance % and money lost. The top shows whether usage is normal and the estimated loss.
4. When items are over the threshold, click "Notify HR" to send the list to HR.

The page also tells you how many POS products are not yet mapped to a recipe. Map them first so the figures are accurate.

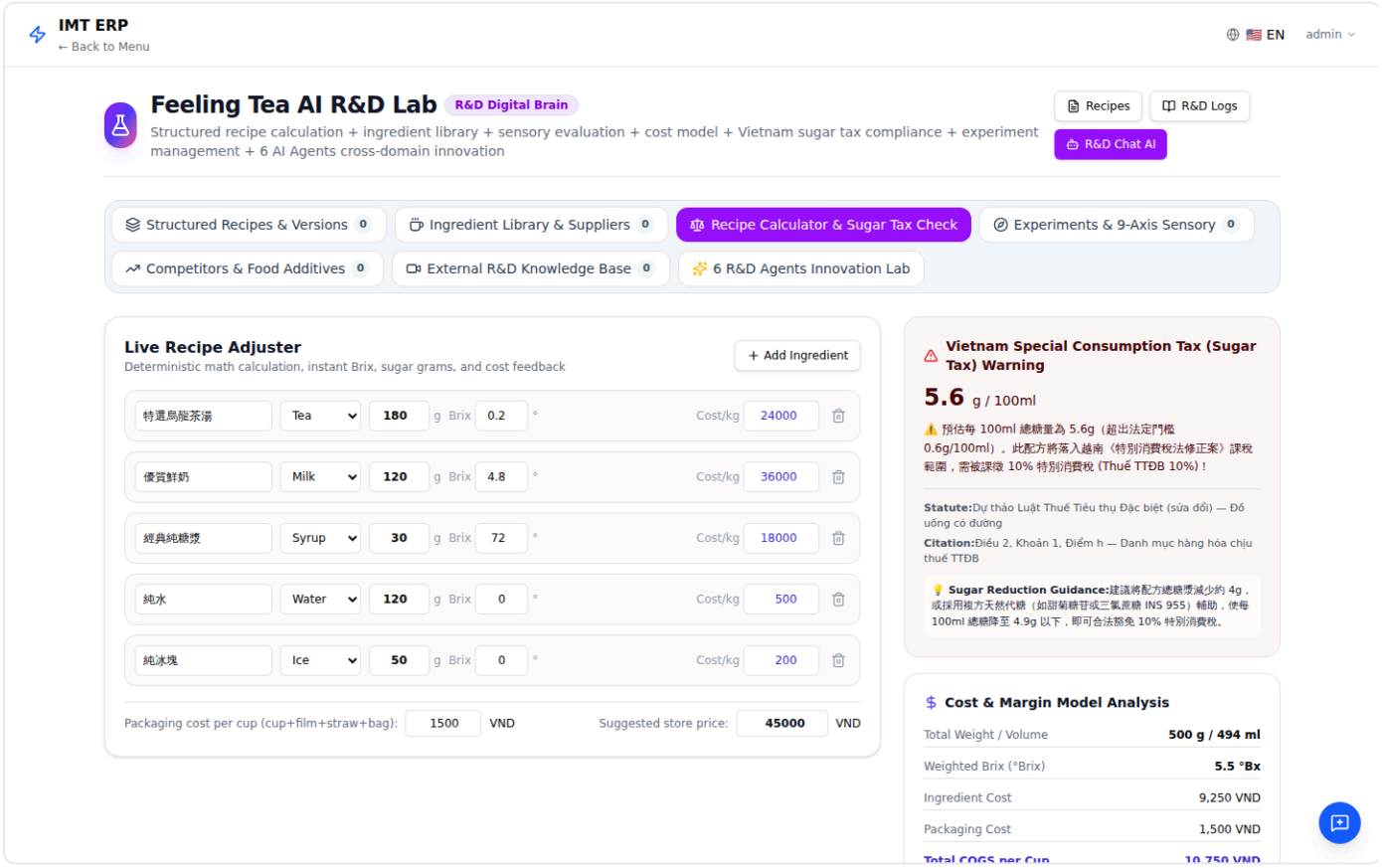
R&D Brain (Lab)

The R&D Brain has structured recipes, calculators and AI tools in seven tabs:

Tab	What it does
Structured Recipes & Versions	Ingredient ratios, volume, sugar, COGS per cup, margin and sugar tax status per product
Ingredient Library & Suppliers	Ingredient cards: supplier, cost, shelf life, sensory scores
Recipe Calculator & Sugar Tax Check	Try out ingredient ratios, sugar, cost and Vietnam sugar tax in real time
Experiments & 9-Axis Sensory	Control vs A/B/C trials, 9-axis sensory scores and conclusions
Competitors & Food Additives	Vietnam market competitor data and food additive (INS) compliance data
External R&D Knowledge Base	Paste a YouTube video, paper, patent or article; AI extracts the technique and rates the evidence
6 R&D Agents Innovation Lab	Six AI roles that help with R&D



Recipe Calculator & Sugar Tax Check



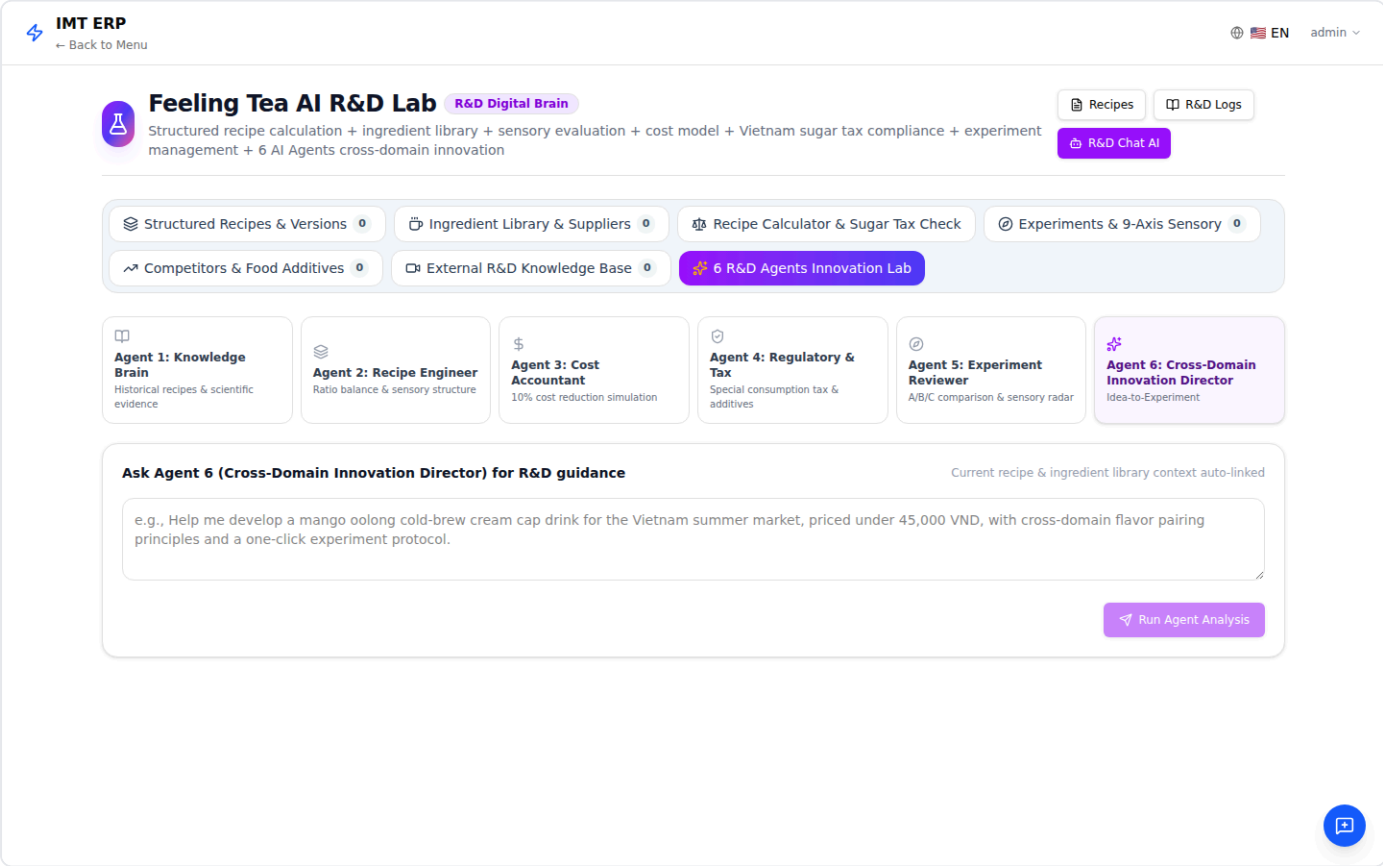
1. Each row is one ingredient: name, category (tea, milk, syrup, water, ice, foam, juice, topping), quantity (g), sugar (°Brix) and cost per kg. Click "Add Ingredient" for a new row.
2. Enter the packaging cost per cup (cup, film, straw, bag) and the suggested store price.
3. The right side shows sugar per 100 ml, total weight, weighted °Brix, ingredient cost, total COGS per cup and gross margin.
4. When sugar is above 5 g/100 ml the system warns that Vietnam's special consumption tax on sugar may apply, with the legal basis and sugar-reduction tips. Check the latest official notices before any tax filing.

The calculator is for trying things out and does not save recipes. Enter the final recipe under "Recipes & Store Cost Calculation".

External R&D Knowledge Base

Paste a YouTube URL, paper DOI or article URL, enter a topic and choose the source type (YouTube, academic paper Level A, food patent Level B, expert article Level C, web article Level D). You can also paste key points from the video or the paper's abstract. Click "Start Learning & Import to Knowledge Base"; AI extracts the key techniques and stores them in the R&D Brain.

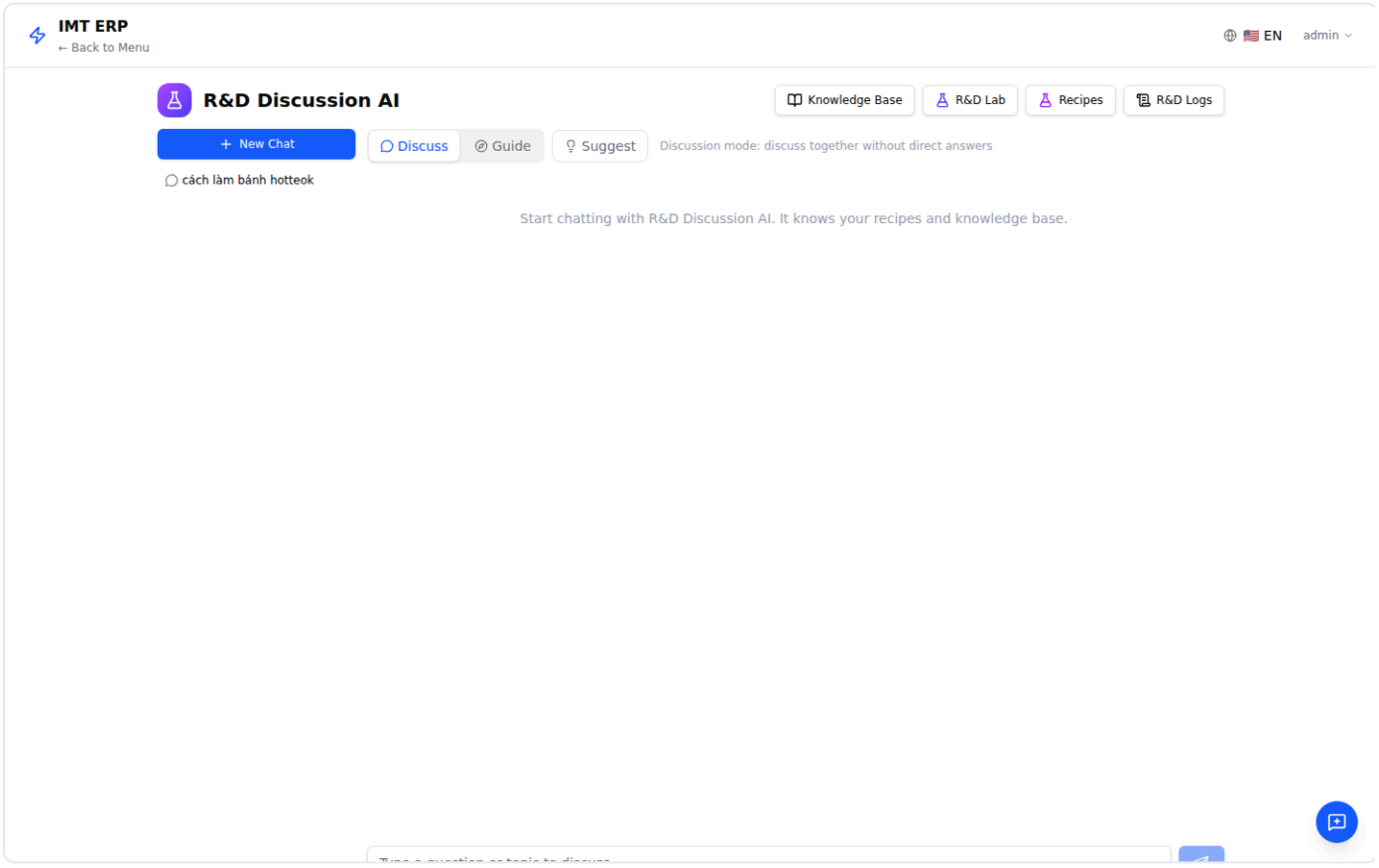
6 R&D Agents



Agent	Focus
1 Knowledge brain	Past recipes and scientific evidence
2 Recipe engineer	Ratio balance and sensory structure
3 Cost analyst	Cutting cost 10% by swapping ingredients
4 Regulation & tax	Special consumption tax and additives
5 Sensory tester	A/B/C comparisons and sensory radar
6 Cross-over innovation director	From idea to experiment protocol (Idea-to-Experiment)

Choose an agent, type your request (for example "develop a mango oolong cold brew with cheese foam for Vietnam's summer, priced under 45,000 VND") and click "Run Agent Analysis". The agent takes the current recipes and ingredient library into account. Test and taste any AI suggestion before using it.

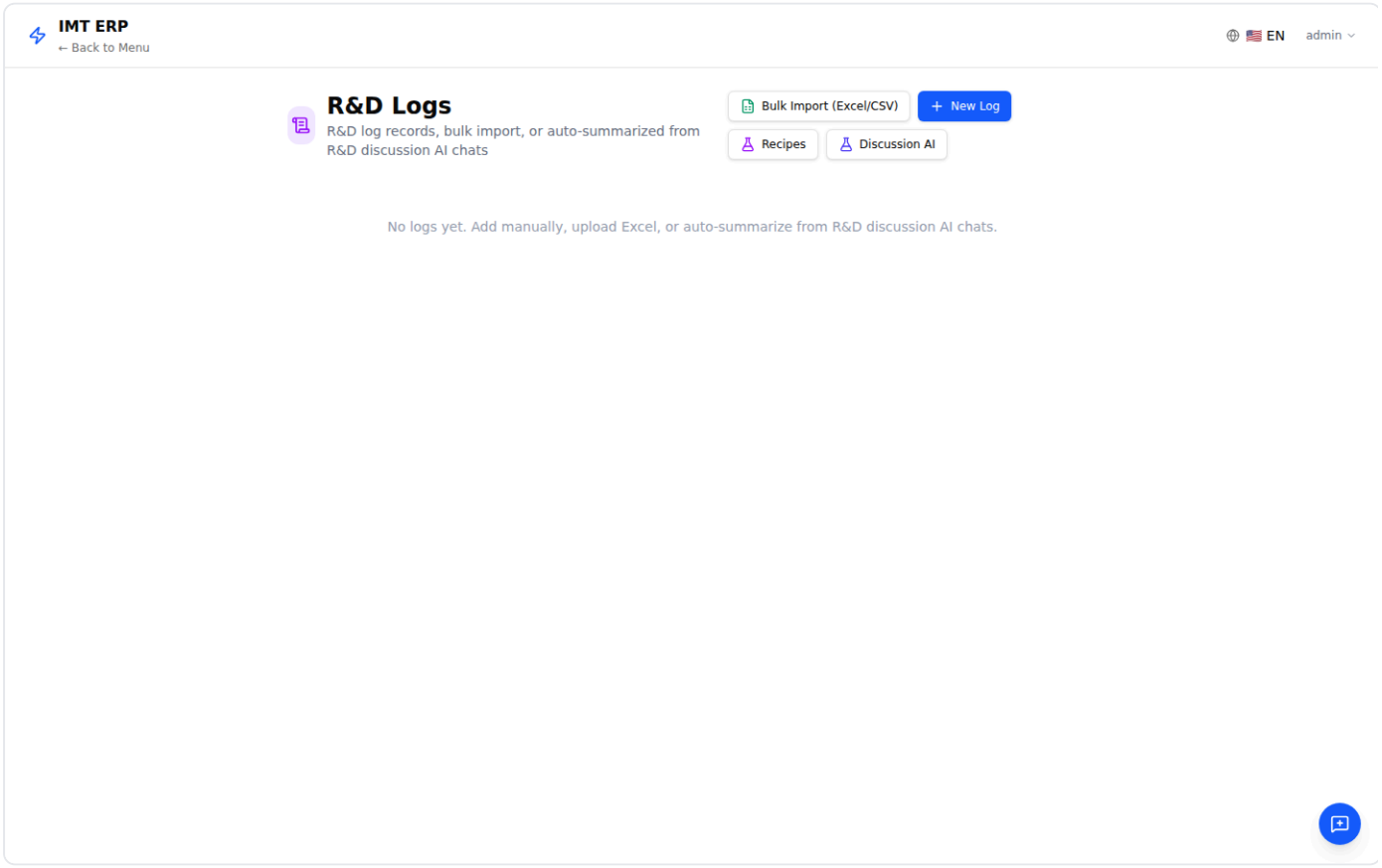
R&D Discussion AI



1. Click "New Chat" on the left; past chats are listed below and can be opened or deleted.
2. Choose a mode:
3. **Discuss**: talk it through together without being handed the answer.
4. **Guide**: AI takes you forward step by step.
5. Turn on "Suggest" to show AI's concrete suggestions in the Suggested Answer panel on the right.
6. Enter sends, Shift+Enter adds a new line.
7. "Knowledge Base" at the top right lets you add material (recipe, company product, external product, note) for AI to use in chats.

Every 4 messages, the system summarises the chat into an R&D log automatically.

R&D Logs



- **New Log**: enter a title and content, then click "Save".
- **Bulk Import (Excel/CSV)**: download the template, fill in the log title and log content, then upload it.
- **Automatic summaries**: R&D Discussion AI chats are turned into logs automatically; each chat keeps updating its own log.

FAQ and notes

Can't open the R&D system? Your account must belong to the R&D unit. Ask an administrator to add R&D to your units.

A recipe cost is 0 or wrong? Check whether an ingredient shows "⚠️ Not priced by Finance yet". Prices are maintained by Finance; ask them to add or correct the price under "Finance Material Pricing" and the recipe cost updates automatically.

Can I change material prices in R&D? No. R&D cannot change or import prices. Contact Finance.

Usage Verification shows no data? POS sales and inventory data for that store and month must be imported, and POS products must be mapped to recipes.

Can I use AI recipe suggestions, sugar tax or regulatory information as is? Not as the sole basis. Recipes must be test-made and tasted, and tax and additive rules must be checked against the latest notices from the Vietnamese authorities.